THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 2: **Electricity: Circuits and their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The reason that coils in heaters are wound is to: 1

(a) look good **(b)** save electricity
(c) fit a long wire in a small space **(d)** give light

2. Which of these is not a good conductor? 1

(a) tap water **(b)** salt water **(c)** distilled water **(d)** lemon juice

3. A short circuit can be prevented by: 1

(a) overloading **(b)** proper insulation of wires
(c) using bare wires **(d)** wet hands

4. A chain of fairy lights is arranged so that one fused bulb causes all lights to go off. This is because the bulbs are in: 1

(a) parallel **(b)** series **(c)** open circuit **(d)** short circuit

5. Which of the following devices converts electrical energy to heat? 1

(a) immersion heater **(b)** bell **(c)** fan **(d)** radio

6. Which of these should be done before cleaning an electrical appliance? 1

(a) switch off the supply and remove the plug
(b) keep the switch on
(c) wet the plug
(d) touch the wires

7. Which of these is a sign of overheating of a socket? 1

(a) sweet smell **(b)** cool feel **(c)** green light **(d)** burning smell

8. A high rating fuse is replaced by a thick copper wire. This is dangerous because: 1

(a) it will not break the circuit on overload
(b) it will not conduct
(c) it will not heat
(d) it will melt too soon

9. A bulb needs the current through its filament because: 1

(a) the current heats it until it glows **(b)** cell makes it cold
(c) air makes it glow **(d)** the filament needs water

10. A fuse is connected in series with an appliance so that: 1

(a) it can light the bulb **(b)** the whole current passes through it

(c) the cell gets charged **(d)** the appliance gets double current

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is it dangerous to fly kites near electric wires? 2

12. Why is a bulb with a broken filament called fused? 2

13. How does a fuse differ from an ordinary wire in a circuit? 2

14. Why is the heating coil of an iron made of an alloy instead of pure copper? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why metals are used for wires and plastic for covers. 3

16. Give three reasons why LED bulbs are preferred. 3

17. Why is it necessary to keep the circuit away from water? 3

OR

Why do electric appliances get warm during use? 3

18. What is the function of a switch in a circuit? Draw the symbol. 3

OR

Explain why a lit bulb is hot to touch but a LED is not very hot. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how batteries of cells are made and calculate the voltage for 3 and 5 cells of 1.5 V. 5

OR

Explain why circuits at home have fuses or MCBs in different ratings for lights, geysers and ACs. 5

20. Compare LED and filament bulb in energy use and heat produced. 5

OR

Describe a circuit with a cell, a bulb, a switch and a fuse and explain the path of current. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND